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General Comment

Subject: Response to Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

We are writing to provide input on developing the Artificial Intelligence (AI) Action Plan, as requested in the Federal Register notice published on February 6, 2025. Our insights are informed by the AI Institute for Resilient Agriculture (AIIRA, see <https://aiira.iastate.edu/>), a collaborative initiative led by Iowa State University with faculty from eight universities and organizations across the United States. AIIRA directly impacts thousands of farmers through its partnership with the Iowa Soybean Association and extensive collaborations with industry stakeholders and the startup ecosystem.

The Role of Higher Education in Advancing AI in Agriculture: Higher education institutions play a critical role in driving AI innovation, particularly in agriculture. Universities serve as centers for interdisciplinary research, integrating expertise in AI, plant sciences, robotics, data analytics, and social sciences to tackle complex agricultural challenges. At AIIRA, this collaborative approach has led to significant developments, including AI-driven predictive digital twins that enhance agricultural resiliency and profitability, as well as foundational AI models that help producers identify and mitigate pests in a timely fashion. AIIRA also provides unbiased assessments of AI tools for industry partners.

AIIRA demonstrates the transformative integration of AI into agriculture through higher education collaboration in the following ways:

Innovative Research: AIIRA is pioneering predictive digital twins, advanced robotics, and robust computer vision models that enable precise, resource-efficient, and resilient agricultural practices.

Interdisciplinary Collaboration: AIIRA unites experts across universities, industry partners, farmers, and startups, leveraging multiple perspectives and expertise to ensure AI innovations are practical, scalable, and impactful.

Educational and Outreach Initiatives: AIIRA delivers training workshops, educational resources, and outreach programs, fostering widespread AI adoption among farmers, commodity groups, and the agricultural workforce. AIIRA also incorporates initiatives focused on upskilling and reskilling the existing workforce within small and medium-sized enterprises (SMEs) across the Midwest to enhance workforce adaptability and ensure that Middle America remains economically competitive in the era of AI-driven advancements.

Recommendations for the AI Action Plan: Drawing from AIIRA's comprehensive vision and community-wide strategic engagements, we recommend the following actions:

- Enhanced Support for Interdisciplinary Research:** Fund collaborative initiatives integrating AI with agriculture (AI+Ag), leveraging academic expertise to overcome barriers such as data scarcity and technology accessibility, benefiting both large-scale operations and smaller farms.
- Investment in AI Education and Workforce Development:** Expand interdisciplinary educational programs that address skills gaps in agriculture, training professionals capable of deploying and utilizing AI technologies effectively across diverse agricultural contexts.
- Establish/Sustain National AI Research Infrastructure:** Create national-scale infrastructure with advanced computing facilities, persistent data storage, and secure data-sharing platforms to enable robust academic AI research and accelerate technology transfer from academia.

to industry and farms.

4) Strengthen Public-Private and Startup Partnerships: Encourage and facilitate strategic collaborations among universities, industry partners, startup ecosystems, and commodity groups like the Iowa Soybean Association, and Iowa Corn Growers Association to expedite the practical deployment and adoption of AI innovations in agriculture.

5) Ensure Practical and Effective AI Deployment: Develop policy guidelines that support AI innovation while addressing legal and societal considerations, ensuring technologies remain transparent, effective, and accessible to all, including rural and agricultural communities, without burdensome restrictions that hinder progress.

By incorporating these recommendations, the AI Action Plan can harness the strengths of higher education institutions and their partnership with industry and agricultural communities to drive significant advancements in AI, bolstering agricultural productivity, economic competitiveness, and sustainability.

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